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The influence of degradation effects in proton exchange membrane fuel cells on life cycle assessment modelling and environmental impact indicators



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HIGHLIGHTS

- Semi-empirical model was developed using performance degradation data of PEMFC.
- New Life Cycle Inventory (LCI) for PEMFC operational phase was defined.
- Operational regime and stack replacement was integrated in newly developed LCI model.
- Environmental impacts of 1kWel PEMFC were analysed.
- Sensitivity analysis for reduced platinum loading and improved durability was performed.

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ABSTRACT

Although proton exchange membrane fuel cell (PEMFC) systems are expected to have lower environmental impacts in the operational phase, compared to conventional energy conversion systems, there are still certain economic, operational, and environmental setbacks. Durability under a wide range of operating conditions presents a challenge because degradation processes affect the PEMFC efficiency. Typically, life cycle assessment (LCA) of PEMFC systems do not include performance degradation. Thus, a novel semi-empirical PEMFC model is developed, which includes degradation effects caused by different operational regimes (dynamic and steady-state). The model is integrated into LCA through life cycle inventory (LCI) to achieve a more realistic and accurate evaluation of environmental impacts. Verification of the model clearly showed that the use of existing LCI models underestimates the environmental impacts. This is especially evident when green hydrogen is used in PEMFC operational phase, where manufacturing phase and maintenance (stack replacements) become more influential. Input parameters of the model can be modified to reflect technological improvements (e.g. platinum loading or durability) and evaluate the effects of future scenarios.

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Abbreviation

ADP elem	Abiotic depletion potential, elements
ADP fossil	Abiotic depletion potential, fossil
AP	Acidification Potential
AST	Accelerated Stress Test
BoL	Beginning-of-Life
CRM's	Critical Raw Materials
EoL	End-of-Life
EP	Eutrophication Potential
EU	European Union
FAETP	Freshwater Aquatic Ecotoxicity Potential
FC	Fuel Cell
FCEV	Fuel Cell Electric Vehicle
FCH JU	Fuel Cells and Hydrogen Joint Undertaking
GWP	Global Warming Potential
HTP	Human Toxicity Potential
LCA	Life Cycle Assessment
LCI	Life Cycle Inventory
LCIA	Life Cycle Impact Assessment
MAETP	Marine Aquatic Ecotoxicity Potential
MEA	Membrane Electrode Assembly
NG-SR	Natural Gas Steam Reforming
ODP	Ozone-Layer Depletion Potential
PEMFC	Proton Exchange Membrane Fuel Cell
POCP	Photochemical Ozone Creation Potential
RES	Renewable Energy Sources
RUL	Remaining Useful Life
TETP	Terrestrial Ecotoxicity Potential

Introduction

Energy and environmental problems are closely related, since it is impossible to generate energy needed in different sectors of society without having any impact on the environment. Paris agreement signed by 195 countries, is another attempt of a global action plan to mitigate climate change [1]. In addition European Union has a very challenging goal of achieving carbon neutrality by 2050 [2]. The major cause of environmental problems is the extensive use of fossil fuels, which provide around 80% of total primary energy demand [3]. Additional challenges in the future energy supply are the limited fossil fuel resources, which according to some forecasts will last at least next 50 years [4]. The need to address climate change is the major driver for a transition from an energy sector dominated by fossil fuels to one based on renewable energy sources (RES) and hydrogen is assumed to be one of the important players in the future energy systems [5]. Hydrogen has several attractive features, beyond the obvious that is the lightest, most efficient, and cleanest fuel that can be converted to electricity by electrochemical processes in fuel cells, and with higher efficiencies than combustion of fossil fuels in internal combustion engines and power plants [6]. Of particulate importance are its useful features of relative transportability (e.g. compared to heat) and long-term storability [7]. Energy storage technologies will play a crucial role in balancing energy flows and ensuring the difference between consumption and generation in the future energy supply. Among the various energy storage mechanisms and technologies, such as pumped storage power plants, batteries, compressed air, flywheels, capacitors, and others, hydrogen technologies will be included in the construction of our future energy system because they enable energy efficient and environmentally sustainable conversion of energy flows between renewable energy sources (RES) and energy consumers [8].

When green hydrogen storage systems are integrated into the overall energy sector, there are multiple opportunities for sector coupling of hydrogen. It can be used in transportation, energy-intensive industry, and the public sector, as shown by H. Gils et al. [9] for the case of Germany, where interaction of hydrogen infrastructures with other sector couplings was studied, and B. Emonts et al. [10], where sector coupling opportunities in road transport were studied.

One of the main parts of the hydrogen storage technologies is the fuel cell system that converts chemically stored energy back to electricity. Among fuel cell technologies Proton-Exchange Membrane Fuel Cell (PEMFC) is one of the most attractive and widespread, due to fast start-up, high power density, lightness, compactness and low operation temperature [11]. The function of PEMFCs is to generate electricity from stored hydrogen produced by RES electricity. Compared to combustion engines, the PEMFC system has no moving parts which allows quiet operation without vibration, noise, less maintenance, higher conversion efficiency, and combined with green hydrogen much lower environmental impacts in operational phase [12]. Although PEMFCs are considered to have lower environmental impacts in the operational phase, there are still several economical, operational and environmental setbacks [13]. In the manufacturing phase the use of expensive and environmentally controversial critical materials, such as platinum or ruthenium in Membrane Electrode Assembly (MEA), has a big influence on the economic and environmental impact indicators [14,15]. Important technical barrier linked with operational phase of PEMFC, which manifests itself as the increase of environmental impacts in manufacturing phase, is the fuel cell durability [16–18].

Durability is the ability of a PEM fuel cell or stack to resist permanent change in performance over time [19]. This is of great importance especially under a wide range of operating conditions that occur in different applications with PEMFC. Various organizations (e.g., US Department of Energy) have proposed many different accelerated stress tests (AST) methods to characterize the two key properties of PEMFC systems that are *performance (characterization curve)* and *durability*. With ASTs method we obtain relevant data about the FC system in a significantly shorter time under controlled conditions. This helps to better understand its performance and durability in real operating conditions without reaching failure or the End-of-Life (EoL) of the stack [20–22]. A maximum of 10% capacity reduction (voltage potential at nominal operating conditions of the system) compared to the values at the Beginning-of-Life (BoL) is defined as EoL, when the core component (stack) must be replaced or at least renewed [23]. In order to promote the wider use and commercialization of PEMFC systems, lifetime targets for PEMFC systems have been set to 60,000–90,000 operating hours for steady-state and

5000 h (8000 h - long-term target) for dynamic operation (automotive applications) of PEMFC systems [24]. Due to different operation regimes (dynamic/steady-state) different degradation mechanisms affect the PEMFC stack, which are manifested in the loss of performance and durability, and consequently in increase of environmental impact indicators [25,26]. The loss of performance due to different degradation mechanisms is evident and has been confirmed within studies using different experimental approaches, where various effects causing degradation processes were investigated including temperature cycling, on-off cycling and power cycling [27–29]. Some authors have tried other methodologies, such as short circuiting, hydrogen pumping, and hot water circulation [20]. The most common PEMFC degradation effects were investigated for different driving regimes in accordance with automotive applications (Fuel Cell Electric Vehicles - FCEVs). The results showed that the fuel cell degradation affected the average fuel economy of the FCEV and several authors proposed additional AST studies for FC durability to understand degradation mechanisms [30]. Some authors have also derived semi-empirical degradation models [31], but those models were almost never introduced into Life Cycle Assessment (LCA) approach to evaluate environmental impacts of the system due to degradation effects or/and due to operating regime of the PEMFC system. Since dynamic operation accelerates the degradation and shortens the PEMFC remaining useful life (RUL), the environmental impacts increase due to lower performance of the system and due to needed replacement or renewal of the stack that reaches EoL faster than in steady-state operation regime. During dynamic operation the load change regime has the greatest impact on operating time, followed by the on/off regime, standby mode, and operation at high loads [32]. In the case of highway driving cycle (lower power fluctuations) overall environmental impacts over the PEMFC lifetime are much lower compared to the urban driving cycle, which is usually more dynamic, and this should be implemented in future LCA studies. Degradation effects in fuel cells were studied mainly using experimental approaches, because the lack of deep understanding of all processes prevents evaluation on the modelling scale [33,34].

Literature review of the LCA studies of PEMFC systems revealed that majority of authors did not consider the effects of performance degradation during the lifetime of the PEMFC in their life cycle inventories (LCI) used for the operational phase that affect also manufacturing and EoL phase [35–37]. There is also no evidence if the authors included in LCIs the operating regime (dynamic or steady state), which clearly affects the intensity of performance degradation and compared to nominal operation decreases remaining useful life of PEMFC. The degradation rates in studies for dynamic [22,33,33,38] and steady-state [32,39,40] operation regimes of PEMFC are shown in Fig. 1 with error bars and ranges included.

The study done by P. Ahmadi and E. Kjeang, that was focused on LCA of PEMFC, is the only work, which considered degradation effects in assessment of environmental impacts [41]. They considered the PEMFC degradation of 3.65 $\eta V/h^{-1}$, which they arbitrarily adopted from the literature. They used this degradation value for PEMFC in FCEV although this

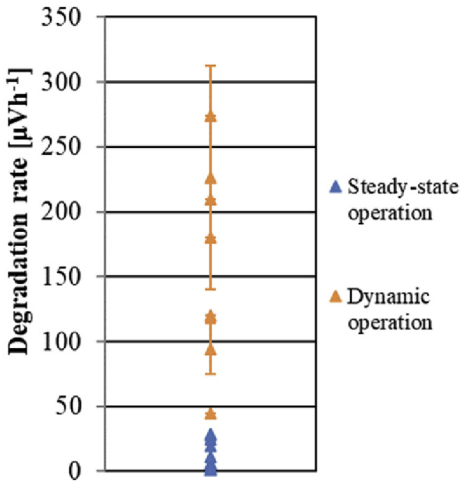


Fig. 1 – Degradation rates ranges obtained from literature review: steady state [32,39,40]; and dynamic [22,33,33,38] operational regime.

degradation rate is more appropriate to use for the PEMFC in steady-state operation regime [42]. Most of the life cycle inventories (LCI) for the PEMFC operational phase are developed without considering the degradation mechanisms and efficiency drop, therefore it is necessary to include degradation processes into LCIs to obtain realistic results and environmental impacts that include performance loss, stack replacements, maintenance, and dependence on operational regime of the PEMFC system.

In this paper a novel workflow of the life cycle inventory development is presented. The semi empirical model obtained from validated experimental data sets of ASTs from literature was integrated into LCI of PEMFC operational phase. This new PEMFC LCI model accounts for the influence of performance loss and durability caused by the operating regime of the PEMFC system. New parameter "operating regime factor" is introduced that enables the user to set the ratio between the dynamic and steady state operating regime of PEMFC, which is usually the mixture of both. With the operating regime factor the LCI model enables a more realistic assessment of environmental impacts of PEMFC in the operating phase. The LCI model is very versatile and fully customizable and enables the assessment of environmental impacts based on input parameters such as energy efficiency of the PEMFC system, rated electric power of the PEMFC system at the BoL, average voltage potential per one MEA, desired operational lifetime, and operating regime factor. The model is developed for operating phase of PEMFC, but it indirectly influences the manufacturing phase also. Based on the inputs and the definition of PEMFC EoL it automatically calculates required number of stack replacements in the entire lifetime (desired life expectancy) of the PEMFC system. If more stacks need to be manufactured in the entire lifetime that influences the environmental impacts in manufacturing phase of the system. In addition, the LCI model is constructed in a way that can be adapted to reflect the technology improvements made in the future.

To evaluate the new LCI model for a PEMFC system a comparative 'cradle to the gate' LCA study of 1 kW PEMFC

system is made. The environmental impacts of the new innovative LCI model (new LCI) are compared to the environmental impacts of currently available LCI model (existing LCI), which does not consider the degradation effects. The sensitivity analysis of different hydrogen production methods, from grey to green hydrogen, helps to emphasise the relevance and relative impact of degradation processes in PEMFC to total environmental impacts. Possible future technology improvements are also evaluated and discussed, such as reduced platinum loading, improved efficiency, and prolonged lifetime targets [43,44].

Methodology

The main purpose of this study is to investigate how the degradation of the PEMFC affects the LCA of environmental impacts, and additionally, to improve the accuracy and flexibility of the modelling. First, it is explained why there is a need to construct a new LCI, and what are the differences and benefits compared to a standard existing LCI. Afterwards, a semi-empirical model, used to assess and compare the degradation impacts, is explained in detail.

Methodological approach and workflow for new PEMFC operational life cycle inventory

Traditionally, the LCA analyses do not consider the degradation of the PEMFC systems in operational phase. Hence, they use nominal operating parameters (e.g. from technical datasheets) throughout the entire lifetime of the system. In reality, this is not the case because degradation effects cause a drop in system efficiency over time, which consequently means higher consumption of fuel for the same required power output. Also, certain components of the system need to be repaired or replaced to allow the system to reach the desired lifetime and optimal operation. All these changes contribute to additional environmental impacts which should not be neglected. To account for these effects a new LCI was constructed as shown in Fig. 2.

As presented in studies [45,46], the lifetime of a PEMFC strongly depends on its operating conditions. To determine and include in the innovative workflow of life cycle inventory the performance degradation of the PEMFC during its lifetime (efficiency and power degradation), ASTs and real data sets from literature were used to gather durability information. Based on that data from literature presented in the following chapter the durability results were analysed and the performance degradation was evaluated. The degradation performance of fuel cell systems was identified and the dependence of the PEMFC system performance and efficiency with respect to time and operating regime was determined. The obtained results were implemented into a new innovative semi-empirical LCI model of PEMFC operation, which is described in more detail in next Section: Semi-empirical PEMFC operation model.

Semi-empirical PEMFC operation model

In this section the modelling approach to constructing the new LCI is explained. First, a mathematical model of mass and energy balance in a PEMFC system is devised. Further on, the operating factor is defined, which accounts for the operating regime of the system (steady-state or dynamic). The boundary values for the operating factor are taken from literature and are based on an extensive study of various ASTs, which aimed at replicating the degradation results of either dynamic or steady-state operation. The model is based on AST data because it is easier to collect the degradation data due to shortened testing time. However, if real data collected from the field operation is available it can also be used in the model. Based on the operational regime (operating factor - F_{op}) of the system the EoL operation time is calculated, which further defines the degradation rate. This degradation rate is then used to define the efficiency drop and increase in hydrogen consumption when the PEMFC system reaches the EoL operation time. Furthermore, the number of stack replacements (N) is calculated if the desired operational lifetime exceeds the calculated EoL time of the PEMFC system.

First step of developing PEMFC operation model is to determine mass and energy balance for PEMFC system, which is shown in Fig. 3. Energy balance is presented in eq. (1). And mass balance in eq. (2).

$$\dot{H}_{H_2(g),R} + \dot{H}_{air(g),R} = P_{el} + \dot{H}_{air(g),P} + \dot{H}_{H_2O(l),P} + \dot{Q}_{out} \quad (1)$$

where: $\dot{H}_{H_2(g),R}$ is enthalpy of reactant Hydrogen, $\dot{H}_{air(g),R}$ is enthalpy of reactant air, $\dot{H}_{air(g),P}$ is enthalpy of produced air, $\dot{H}_{H_2O(l),P}$ is enthalpy of produced water, P_{el} is electric power output of PEMFC and $\dot{Q}_{out}$ is heat released (as losses in surroundings).

$$\dot{m}_{H_2(g),R} + \dot{m}_{air(g),R} = \dot{m}_{air(g),P} + \dot{m}_{H_2O(l),P} + \dot{m}_{H_2,loss} \quad (2)$$

where in mass balance: $\dot{m}_{H_2(g),R}$ is inlet mass flow of H_2 defined in eq. (3)., $\dot{m}_{air(g),R}$ is inlet mass flow of air defined in eq. (4)., $\dot{m}_{air(g),P}$ is outlet mass flow of air and is defined in eq. (5)., $\dot{m}_{H_2O(l),P}$ is outlet mass flow of water and is defined in eq. (6)., and $\dot{m}_{H_2,loss}$ is unused hydrogen at the outlet of PEMFC system and in our model is set to 0 (as we assume no hydrogen outflow due to circulation). In addition, the parameter λ_{H_2} represents the excess hydrogen (above stoichiometric ratio) and if needed can include hydrogen losses or open-ended operation. In our model is set to 1, because no hydrogen losses and recirculation of unused hydrogen are assumed.

$$\dot{m}_{H_2(g),R} = \frac{M_{H_2} \cdot I \cdot \lambda_{H_2}}{2 \cdot F} = 1,05 \cdot 10^{-8} \cdot \frac{P_{el}}{U_{PEMFC}} \cdot \lambda_{H_2} \left[\frac{kg}{s}\right] \quad (3)$$

$$\dot{m}_{air(g),R} = \frac{M_{air} \cdot I \cdot \lambda_{air}}{w_{O_2} \cdot 4 \cdot F} = \frac{28,97 \cdot 10^{-3} \cdot I \cdot \lambda_{air}}{0,21 \cdot 4 \cdot F} = 3,57 \cdot 10^{-7} \cdot \frac{P_{el}}{U_{PEMFC}} \cdot \lambda_{air} \left[\frac{kg}{s}\right] \quad (4)$$

$$\dot{m}_{air(g),P} = \dot{m}_{air(g),R} - \dot{m}_{O_2} \Rightarrow \dot{m}_{air(g),P} = \left(3,57 \cdot 10^{-7} \cdot \lambda_{air} - 8,29 \cdot 10^{-8}\right) \cdot \frac{P_{el}}{U_{PEMFC}} \left[\frac{kg}{s}\right] \quad (5)$$

$$\dot{m}_{H_2O(l),P} = \frac{M_{H_2O} \cdot I}{2 \cdot F} = 9,34 \cdot 10^{-8} \cdot \frac{P_{el}}{U_{PEMFC}} \left[\frac{kg}{s}\right] \quad (6)$$

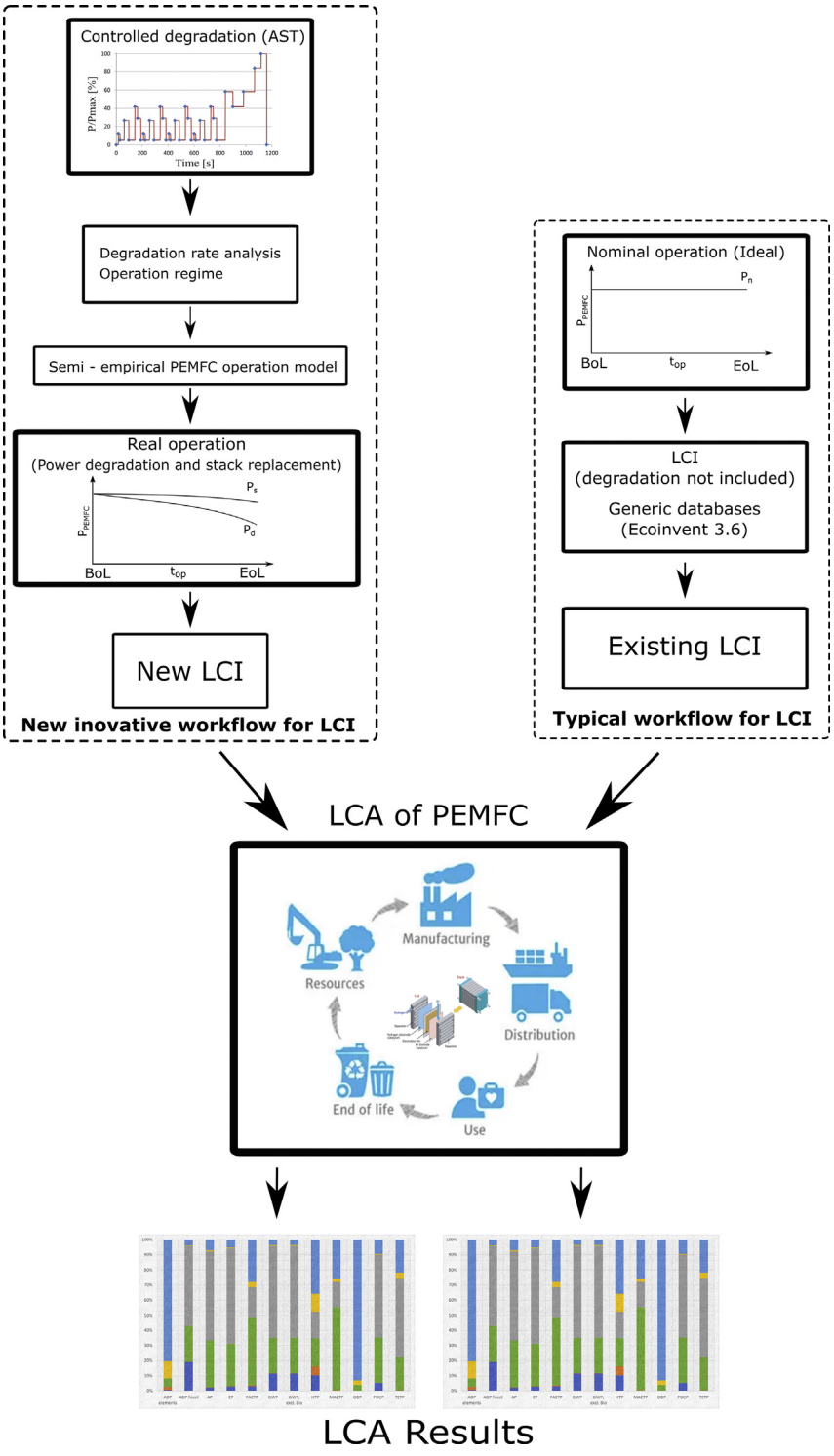


Fig. 2 – Innovative approach and workflow for new life cycle inventory (LCI).

From the energy and mass balance the energy efficiency of the PEMFC system was derived, which is a key parameter for defining the specific consumption of hydrogen ($\dot{m}_{spec_{H_2}}$), which defines needed mass flow of hydrogen ($\dot{m}_{H_2(g),R}$) to produce 1 kW h of electricity with 1kWel PEMFC system. The specific hydrogen consumption is a final parameter for constructing the life cycle inventory (LCI) of the PEMFC operational phase, as directly reflects the PEMFC system efficiency. Usually, PEMFC system manufacturers report energy efficiencies for the lower hydrogen heating value to reflect higher PEMFC energy efficiency. In the presented model, we used the higher heating value of the hydrogen $H_{H_2,HHV}$ (39.4 kW h/kg). In real applications, unused hydrogen at the outlet is usually recirculated back to the inlet, which at the end improves efficiency

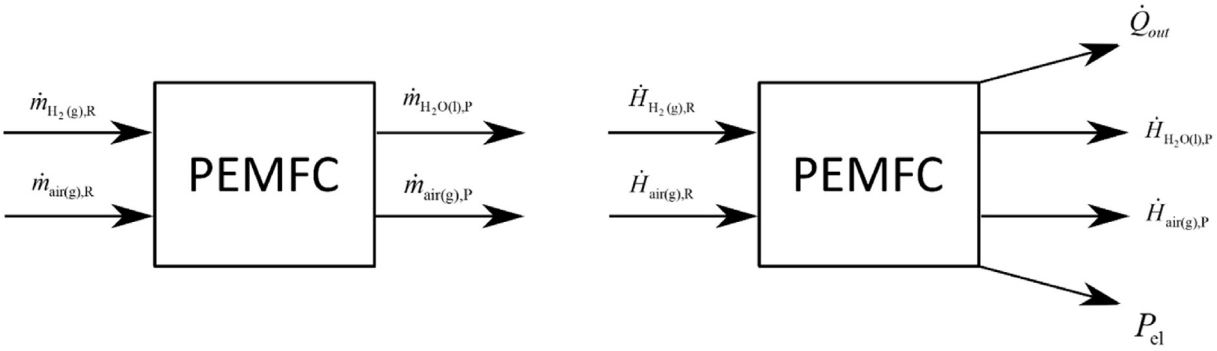


Fig. 3 – Mass (left) and energy balance (right) of PEMFC.

due to higher utilisation of hydrogen in the PEMFC system. In this study, the assumption is made that system operates with negligible hydrogen losses, but the new model allows to include hydrogen losses ($\dot{m}_{H_2,loss}$) if the data is available. Specific hydrogen consumption is the required amount of hydrogen mass flow for 1 kW h produced electricity with the PEMFC system and it is defined as:

$$\eta_{PEMFC} = \frac{U_{PEMFC} \cdot I}{\dot{m}_{H_2} \cdot H_{H_2,HHV}} = \frac{P_{el}}{\dot{m}_{H_2} \cdot H_{H_2,HHV}} \Rightarrow \Rightarrow \dot{m}_{spec_{H_2}} = \frac{1}{\eta_{PEMFC} \cdot H_{H_2,HHV}} \left[\frac{kg_{H_2}}{kWh}\right] \quad (7)$$

The generated heat and the produced electricity were determined using eq. (8). And eq. (9).:

$$\dot{Q}_{out} = \dot{m}_{H_2} \cdot H_{H_{2,HHV}} - P_{el} \quad (8)$$

$$W_{el} = n \cdot I \cdot U_{MEA} \cdot t_{op} = I \cdot U_{PEMFC} \cdot t_{op} = P_{el} \cdot t_{op} \quad (9)$$

where: n is the number of membrane electrode assemblies (MEA) in the PEMFC system, I is the electric current of the PEMFC system, U_{MEA} is the voltage potential of one PEMFC cell, U_{PEMFC} is the voltage potential of the PEMFC system, $H_{H_2,HHV}$ is the higher heating value of Hydrogen and t_{op} is operational lifetime of the PEMFC system.

The LCI results for operation were presented in the form to the required hydrogen consumption for 1 kW h of generated electricity with the PEMFC system. The developed generalized life cycle inventory (LCI) of PEMFC operation is used for hydrogen technology systems with PEMFC for the needs of LCA analysis.

Since the operational lifetime is very important for a more accurate evaluation with the LCA method, as it is crucially influenced by degradation processes, we determined the operational lifetime of the PEMFC by introducing a degradation rate of the voltage potential. The end-of-life is reached when a 10% drop of the voltage potential (EoL) relative to the initial nominal voltage potential of the PEMFC (BoL) occurs [23]. The degradation rate is defined as:

$$D_{PEMFC} = \frac{U_{BoL} - U_{EoL}}{t_{op}} = \frac{0.9 \cdot U_{BoL}}{t_{op}} \left[\frac{\mu V}{h}\right] \quad (10)$$

The average degradation rates, together with the average operational lifetime, were adopted from G. Saur et al. [47] for different fuel cell applications (over 260 datasets collected up to and including year 2018). Based on the average operational lifetime (t_{op}) of the PEMFC system at a 10% voltage drop (EoL), an average operational lifetime of 3800 h for dynamic and 11,900 h for stationary applications are assumed in the presented LCI PEMFC operation model. According to eq. (10). The average degradation rate at nominal (BoL) fuel cell voltage of 0.6 V is 16 μV h^{-1} for 3800 h of dynamic operation and 5 μV h^{-1} for 11,900 h of stationary operation. The results of PEMFC degradation rate as a function of PEMFC operational lifetime for 3 different nominal fuel cell voltage potentials (0.5, 0.6 and 0.7 V) on a single PEMFC are shown in Fig. 4.

To model the influence of the operating regime and the degradation rate on the operational lifetime of the PEMFC, the operating factor (F_{op}) was introduced into model. The F_{op} defines the number of operating hours until 10% of the voltage drop occurs, which is defined as the EoL of the PEMFC system. Values of the operating factor F_{op} range from 0 (steady-state mode of operation) to 1 (dynamic mode of operation). The semi empirical LCI model allows the operating factor input values to be between 0 and 1 (e.g., 0.2 for PEMFC system with 20% of time is under dynamic operation). This value corelates to a new operational lifetime (t_{op}) and degradation rate of the PEMFC system, which are automatically calculated using eq. (10). And eq. (12). F_{op} boundaries are defined as:

$$F_{ob} = \begin{cases} 1 \Rightarrow t_{EoL,dynamic} = 3,800 \text{ h} \\ 0 \Rightarrow t_{EoL,stationary} = 11,900 \text{ h} \end{cases} \quad (11)$$

The benefit of this life cycle inventory model is that the presented values for EoL time and degradation rate can be continuously updated as the PEMFC technology state of the art of core components evolves and improves. In the LCI operational model, the operational lifetime of the PEMFC system is defined as:

$$t_{op,EoL} = -F_{op} \cdot (t_{EoL,stationary} - t_{EoL,dynamic}) + t_{EoL,stationary} \quad (12)$$

where $t_{EoL,steady-state}$ is average operational lifetime for steady-state operation of the PEMFC system and $t_{EoL,dynamic}$ is average operational lifetime for dynamic operation of the PEMFC system.

One of the main parameters in LCI PEMFC operational model is drop in energy efficiency, which consequently increases the specific hydrogen consumption. The energy

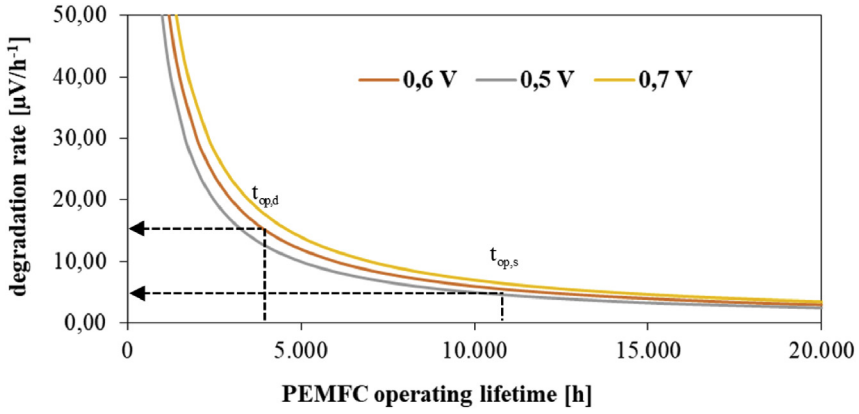


Fig. 4 – Degradation rate depending on PEMFC operational lifetime for 3 different rated voltages.

efficiency of the PEMFC and specific hydrogen consumption at EoL is defined as:

$$\eta_{BoL} = \frac{I \cdot U_{PEMFC_{BoL}}}{\dot{m}_{H_2} \cdot H_{H_2,HHV}} = \frac{P_{el}}{\dot{m}_{H_2} \cdot H_{H_2,HHV}} \Rightarrow \Rightarrow \dot{m}_{spec,BoL_{H2}} = \frac{P_{el}}{\eta_{BoL} \cdot H_{H_2,HHV}} \tag{13}$$

$$\eta_{EoL} = \frac{I\left(U_{PEMFC_{BoL}} - t_{op} \cdot D_{PEMFC}\right)}{\dot{m}_{H_2} \cdot H_{H_2,HHV}} = \frac{I \cdot U_{PEMFC_{EoL}}}{\dot{m}_{H_2} \cdot H_{H_2,HHV}} \Rightarrow \Rightarrow \dot{m}_{spec,EoL_{H2}} = \frac{P_{el}}{\eta_{EoL} \cdot H_{H_2,HHV}} \tag{14}$$

For dynamic operation of the PEMFC the average degradation rate of energy efficiency is 0.88% per 1000 h of operation, while for steady-state operation the degradation rate of PEMFC efficiency is 0.28% per 1000 h. These reference values (η_{BoL} = 0.4) are incorporated in our model and refer to the upper heating value of hydrogen and the reference degradation rates presented above at a nominal voltage of 0.6 V per one cell. These values and eq. 2.-14. Were used to develop the new semi-empirical LCI operating model, in which degradation effects in the operating phase of the PEMFC system are considered. In addition, a new parameter "N" has been introduced in the PEMFC life cycle inventory model, which defines the number of PEMFC stack replacements during the whole operational lifetime of the PEMFC system. Depending on the operating regime, where the index N_s represents steady-state operation and N_d represents the dynamic operation. N is defined as (see Fig. 5):

$$N_i = \frac{t_{op}}{t_{op,EoL}} \tag{15}$$

This new implemented parameter N indirectly affects the manufacturing phase and the EoL phase of the PEMFC system. With this parameter, the LCI model presented is even more accurate/complete and influences the LCA analysis in other life cycle phases, namely the environmental impacts related to the EoL and consequently effects manufacturing and operational phase of the PEMFC systems. Due to the degradation processes, the energy efficiency of the PEMFC system decreases during the operational phase, which means that the hydrogen consumption increases to provide the same power output of the PEMFC system under the given operating conditions. Lower PEMFC lifetime and higher hydrogen consumption (durability and performance) translate into higher environmental impacts during hydrogen production, PEMFC stack manufacturing and end-of-life phase. With degradation the presented model emphasise and include the importance of critical materials (PGM materials) contribution to the environmental impact in the entire life cycle of the PEMFC system [47–49].

The main input parameters in LCI model are.

1) energy efficiency (η_{BoL}) of the PEMFC system at the beginning of operation (BoL),
2) rated electric power (P_{el}) of the PEMFC system at the BoL,
3) average voltage potential (U_{PEMFC}) at one MEA of PEMFC system,
4) Desired operational lifetime (t_{op}) and
5) operational factor (F_{op}).

With operating factor (F_{op}) the operational regime, operating time until EoL and the degradation of the PEMFC energy efficiency are defined. Additionally, indirectly F_{op} defines the required number of PEMFC stack replacements (N parameter) and thus includes additional environmental impacts due to the production of a new PEMFC stack. The presented LCI model is prepared in a way that when the guidelines and standards for EoL strategies [50,51] for core PEMFC components will be defined they can be included within the model.

Life cycle assessment approach

Life cycle assessment approach used in this study follows the four standard steps of LCA, which are: goal and scope definition, life cycle inventory (LCI) analysis, life-cycle impact assessment (LCIA), and interpretation of the results. The research was conducted according to the ISO standards 14,040, 14,044 [52,53], and International reference Life Cycle Data System (ILCD) handbook [54]. The suggestions given in the guidance document from EU project FC-HyGuide for performing a LCA on Fuel Cell and Hydrogen technologies, were

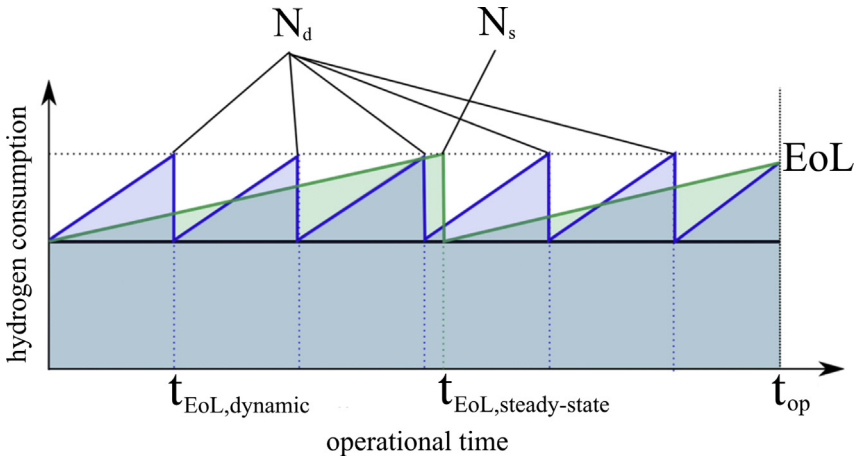


Fig. 5 – Hydrogen consumption and the number of stack replacements depending on operating regime and EoL.

also considered [55,56]. GaBi software environment with integrated Ecoinvent database was used to model and assemble all LCA models included in this study [57].

Goal and the scope of the study

The main goal of the LCA study was to evaluate the LCI model, presented in the previous section, that was integrated into the LCA software to assess environmental impacts of degradation effects in the operational phase of the PEMFC system. Environmental impacts obtained with the new LCI model were compared to those obtained with generic LCI model in GaBi software that does not include degradation effects. The functional unit (FU) used in the study was 1 kW h of electricity generated by the 1 kWel PEMFC system. The scope of the study was cradle to gate, which includes all processes from the raw material extraction, manufacturing phase of the PEMFC system, sensitivity analysis of different hydrogen production technologies and maintenance with needed stack replacements to the of end of the PEMFC system operation. Reference flow of the study is 20,000 h of the 1 kWel PEMFC system operation in the entire scope of the analysis. This is the lifetime that balance of plant components can withstand without major issues, but stack must be replaced several times depending on operational regime.

Life cycle inventory analysis (LCI)

Manufacturing phase

In the manufacturing phase of the PEMFC system, the inventory for fuel cell and balance of plant components from preliminary studies was used [58]. The input data for the material and energy flows was obtained from Ecoinvent 3.5 and the EU funded project NEEDS [32]. In Table 1 inventory for manufacturing phase is presented in the form of mass and energy balances for 1 kWel PEMFC system. In addition, the processes from the generic LCI databases Gabi Professional and Ecoinvent 3.5., that were used for the LCA model, are listed. The manufacturing phase is schematically shown also in Fig. 6. The LCI for the manufacturing phase is somewhat outdated, but it serves the purpose of LCI model evaluation since the comparison of degradation effects is done on the same system. Since the LCI model is open source, it enables changing different parameters that are linked with future technological progress. In the base reference case (Table 1) platinum (Pt) loading is 0.75 g of Pt per 1 kWe power of PEMFC [48]. Additional analysis for Pt loadings of 0.4 g kWe^{-1} and 0.125 g kWe^{-1} was done according to targets set by EU [59] and US Department of Energy [60,61]. Electricity consumption from the EU 28 energy mix was assumed for the energy use in the manufacturing phase of the components and for the final assembly of the 1 kWel PEMFC system. All mass and energy balances for manufacturing phase are normalized to 1 kWe power of the PEMFC system.

Operational phase

The novelty in this study is the LCI model of PEMFC that was developed to assess the influence of degradation effects of the PEMFC stack on the environmental impacts in the operational phase. All options considered in the operational phase are presented in Fig. 6. The fuel for PEMFC is hydrogen that can be produced with different methods which have different environmental impacts. Degradation effects, which are the focus of the research, are reducing the efficiency of the PEMFC over time, and consequently hydrogen consumption is increased to sustain the same power output. When the efficiency of the PEMFC system drops to the point where the fuel cell stack voltage level is lower than 90% of the initial (nominal) voltage, which that is considered as end of life (EoL) for a stack, the stack must be replaced with a new one [13] to maintain the recommended PEMFC system performance. Two main subphases - of the operational phase considered in LCA study - are hydrogen production and maintenance that includes also stack replacements.

To evaluate the developed LCI model we compared environmental impacts of the existing LCI model in Ecoinvent 3.5 database and the new developed LCI model. To analyse the differences of LCI models two extreme operational regimes were considered: steady state and dynamic operational regime, where there is significant difference in degradation rate and stack replacements. The operational factors used were $F_{op} = 0$ for steady state and $F_{op} = 1$ for the dynamic mode of operation. The power of the PEMFC system considered was 1 kWel with the efficiency at the beginning of life (η_{BoL}) 40%,

Table 1 – Life cycle inventory for the manufacturing phase of 1 kWe PEMFC system [48].

	Material	Unit	Value	Manufacturing process used
Stack	Graphite	kg	4.5	GLO: market for graphite ecoinvent 3.5
	Polyvinylidene chloride (PVdC)	kg	1.1	RER: Polyvinylidene chloride (PVdC) PlasticsEurope
	Aluminium	kg	0.3	EU-28: Aluminium ingot mix ts
	Chromium steel	kg	0.1	RER: steel production, electric, chromium steel 18/8 ecoinvent 3.5
	Glass fibres	kg	0.1	DE: Glass fibres ts
	Perfluorosulphonic acid (PFSA) (Nafion®)	kg	0.07	CA: Nafion - for use in fuel cell ts
	Carbon black	kg	0.0008	GLO: market for carbon black ecoinvent 3.3
	Platinum	kg	0.00075	GLO: market for platinum ecoinvent 3.3
BoP	Steel product	kg	3.7	DE: EAF Steel billet/Slab/Bloom ts
	Polyethylene high density granulate (HDPE)	kg	1.5	GLO: market for polyethylene, high density, granulate ecoinvent 3.3
	Chromium steel	kg	1.1	RER: steel production, electric, chromium steel 18/8 ecoinvent 3.5
	Cast-iron component	kg	0.8	DE: Cast iron component (EN15804 A1-A3) ts
	Aluminium	kg	0.75	EU-28: Aluminium ingot mix ts
	Polypropylene granulate (PP)	kg	0.25	GLO: market for polypropylene, granulate ecoinvent 3.3
System	Electricity	kWh	16.9	EU-28: Electricity grid mix ts

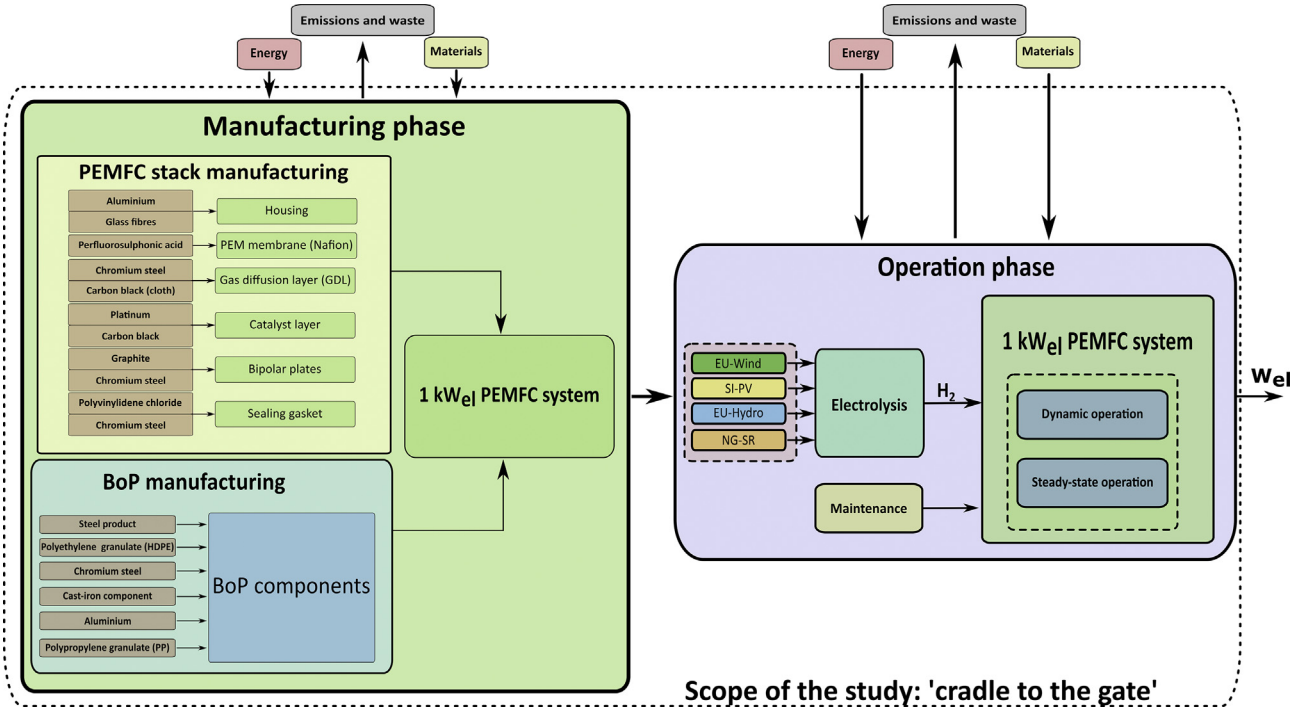


Fig. 6 – Cradle to the gate LCA study for 1 kW PEMFC system.

which corresponds to the average typical PEMFC system efficiency (for high heating value of hydrogen) [62–64]. Operational time was set to 20,000 h at power of 1 kWe. Due to the degradation effects the hydrogen consumption must be increased to always generate the power of 1 kWe. In the case of steady state operation one replacement ($N_s = 1$) of the stack is included (eq. (15).), and in the case of dynamic operational regime 5 replacements ($N_d = 5$) of the PEMFC stack are included.

Hydrogen production

To evaluate the environmental importance of hydrogen production in the future energy systems two different hydrogen production methodologies were included in the study: natural gas steam reforming and electrolysis. In 2020 the majority (~95%) of hydrogen was still produced from fossil fuels [65–69]. As the baseline scenario for hydrogen production the natural gas steam reforming (NG-SR) was used, which is the most widespread hydrogen production technology and represents more than 48% of total hydrogen produced, followed by oil reforming (30%) coal gasification (18%) and only 4% represents other technologies with water electrolysis included [69,70]. To reduce environmental footprint of the produced hydrogen electrolysis powered by electricity from RES must be introduced, which is currently on top priority of investments in the EU energy sector [71,72]. In the present study electricity from EU28 Hydro power (EU-Hydro), EU28 wind power (EU-Wind) and SI PV photovoltaic powerplants (SI-PV) were considered and environmental impacts of green hydrogen produced with electrolysis were benchmarked to natural gas (NG-SR) steam reforming case. With this approach the importance of hydrogen production methodology - that

must meet certain minimum environmental criteria – was addressed. This allows to further emphasise and discuss the environmental impacts arising from degradation effects. The LCA model for electrolysis was set-up based on theoretical approach with 4.718 kWh_{el} for production of 1 Nm^3 of hydrogen (1 atm and 0 °C). The efficiency of the electrolyser stack is defined as 0.71 and the efficiency of the overall electrolyser system 0.63 [73,74]. For the hydrogen production the electrolyser manufacturing was excluded from this study, only electricity generation and energy balances in the process of electrolysis are included in hydrogen production. The output results of the LCI model that are inputs for LCA model are the mass and energy flows presented in Table 2.

Maintenance

Maintenance in the operational phase was included as the replacement of the PEMFC stack when it reached its EoL. The inventory from the available database Ecoinvent 3.5 was analysed to define LCI for maintenance in the operational phase [75]. It was found out that Ecoinvent inventory for the PEMFC system maintenance contains outdated and partly deficient data, with no replacements of the PEMFC stack due to degradation effects, and with constant hydrogen consumption throughout the entire modelled lifetime. The number of replacements strongly depends on the operational regime of PEMFC that can be dynamic (e.g. automotive applications), steady-state or combination of both.

In addition to scenario analysis of Pt loading per 1 kWe power in manufacturing phase, the effect of mid-term and long-term durability goals on environmental impacts were tested. According to EU FCH JU defined goals for the PEMFC long-term durability is 8000 h for dynamic and 80,000 h for steady-state operational regime for the year 2050 [59]. In the new LCI model that means 3 stack replacements less in dynamic ($N_d = 2$) and no replacement needed for steady state ($N_s = 0$) operational regime.

Life Cycle Impact Assessment

In the Life Cycle Impact Assessment phase (LCIA), the CML 2001 methodology was used to assess the environmental impacts of the PEMFC for predefined scenarios described in LCI section. CML2001 methodology is widely used therefore the results can be compared with literature in which authors analyse FCH technologies using this LCIA methodology [14,15,47,76,77]. All environmental impact indicators (Table 3) were used to analyse environmental impacts [78].

Table 2 – Life cycle inventory results of 1 kWel power PEMFC for 20,000 h operational time without and with degradation impacts.

Mass and energy flows	existing LCI (Without degradation)	new LCI (With degradation)
	Inputs	Inputs
Hydrogen [kg]	1269	1327
Air [kg]	86,294	90,215
	Outputs	*Outputs*
Electric energy [kWh	**20,000**	**20,000**
Water [kg]	11,288	11,801
Used air [kg]	76,275	79,741
Heat [MJ]	108,000	116,178

Interpretation

Both phases (manufacturing and operational) in the life cycle of the PEMFC system were modelled separately, which enables the assessment of each phase separately with included sub-models (PEMFC stack, BoP and hydrogen production). To be able to make a detailed interpretation of the environmental impacts, the PEMFC system is modelled as a nesting of subplans for each component to easily incorporate the necessary replacement of each component during the operating time of the system. The results are presented in the form of tables with absolute and relative values for all environmental indicators (Table 3) for manufacturing and operational phase summed together. Calculated environmental impacts based on the new developed LCI model - with four different hydrogen production pathways and two operational regimes (steady state, dynamic) - are compared to environmental impacts based on existing LCI model (no degradation effects included) in Ecoinvent 3.5 database. Global warming potential (GWP) indicator is analysed and commented more in details because GWP is one of the main indicators and drivers of EU policies for reducing net greenhouse gas emissions [2]. As mentioned before, the influence of platinum loading and durability on environmental impacts is additionally analysed to evaluate the feasible future scenarios of PEMFC environmental impacts when the technology will be further improved (i.e. lower platinum loading and longer durability).

Results and discussion

In this section, the results of environmental assessment of 1 kW PEMFC system are presented to: (i) verify the new developed LCI model with integrated operational factor regime; (ii) evaluate the relative influence of degradation effects; (iii) analyse the influence of stack replacement on total environmental impacts; and to (iv) make scenario analysis assuming future potential technology improvements, such as lower Pt loading and prolonged durability. The LCA was done using the approach and boundary conditions presented in the previous section. In Table 4 all environmental indicators in absolute values are presented. Furthermore, the global warming potential (GWP) is brought to the forefront and discussed more in details as it is the most common environmental impact indicator used in relevant EU legislations, [2]. The results are shown summarized for manufacturing and operational phase (M + O) with three different scenarios of green hydrogen production with electrolysis (EU Wind, SI PV and EU Hydro). As a benchmark hydrogen produced from natural gas steam reforming (NG-SR) is used.

The influence of operational regime on environmental impacts

To evaluate the newly developed LCI model with included degradation (New LCI) the environmental impacts obtained

Table 3 – CML2001 environmental impact indicators and the areas of protection, [79].

	Abbr.	Name	Indicator	Units
Air/Climate	GWP	Global Warming Potential	Greenhouse gas emissions	kg CO_2 -eq.
	AP	Acidification Potential	Air pollution	kg SO_2-eq.
	POCP	Photochemical Ozone Creation Potential	Air pollution	kg Ethene-eq.
	ODP	Ozone-Layer Depletion Potential	Air pollution	kg R11-eq.
	HTP	Human Toxicity Potential	Toxicity	kg 1,4-DCB-eq.
Water	FAETP	Freshwater Aquatic Ecotoxicity Potential	Water pollution	kg 1,4-DCB-eq.
	MAETP	Marine Aquatic Ecotoxicity Potential	Water pollution	kg 1,4-DCB-eq.
	EP	Eutrophication Potential	Water pollution	kg PO_4-eq.
Soil	TETP	Terrestrial Ecotoxicity Potential	Soil degradation and contamination	kg 1,4-DCB-eq.
Resources	ADP elem.	Abiotic depletion potential, elements	Resource depletion	kg Sb-eq.
	ADP fossil	Abiotic depletion potential, fossil	Resource depletion	MJ

are compared with environmental impacts of the LCI model without degradation (Existing LCI) using different hydrogen production methods. The first column, for each hydrogen production methodology in Table 4, represents results using the existing LCI and in next two columns the results using the newly developed LCI model are presented for steady-state and dynamic operational regime.

The newly developed LCI considers the degradation, and consequently the reduction of the PEMFC energy efficiency, in the operational phase. That results in higher hydrogen consumption and shorter operational lifetime of the PEMFC stack. The PEMFC stack reaches the EoL faster due to performance drop that is dependent on the operational regime. That results in higher environmental impacts as it is presented in Table 4 and S-Table 1. To provide additional information, all the environmental impact indicators are normalized within the same hydrogen production scenario to the values of the impact indicators of the existing LCI (represented as 100%) in S-Table 1.

Results show that there is an expected difference when using different hydrogen production methodologies and there is a clear difference in applying the new LCI model in LCA approach. Using the hydrogen produced with the electricity from the EU Hydro scenario has the lowest environmental impact, followed by the scenario using the hydrogen produced with the wind electricity (EU Wind), and then by PV electricity (SI PV), while scenario using hydrogen produced from NG-SR (grey hydrogen) has the biggest environmental impacts. However, in the case of some environmental impact indicators (e.g., ADP elements) the SI PV scenario is noticeably worse than NG-SR, because of the manufacturing phase of PV cells which includes several materials and pastes with very large environmental impact.

Fig. 7 shows GWP values for all observed H_2 scenarios for manufacturing and operational phase. In addition, the relative increase to the baseline within each H_2 production methodology is listed in the brackets. In the case of hydrogen produced from wind electricity the GWP per 1 kW h increase by 63% (for 24 g CO_2 eq./kWh) in the case of dynamic operation, and 16% (for 6 g CO_2 eq./kWh) in the case of steady-state operation relative to baseline case using existing LCI model. The highest relative increase is in the case of hydro scenario, which is 88% (for 23 g CO_2 eq./kWh) for dynamic, and 20% (for 5 g CO_2 eq./kWh) for steady state operation. The relative increase is smaller in the case of PV (6% for dynamic and 14% for steady state operational regime) and the smallest in the case of hydrogen production from the NG-SR (5% for dynamic and 8% for steady state operational regime). But in absolute terms the emissions in NG-SR scenario are the highest and for dynamic operation the GWP is increased 8% relative to the baseline, which is 52 gCO_2 eq./kWh in absolute value. It is clearly indicated that the degradation has a noticeable relative effect on the increase of the environmental impacts when green hydrogen is used. However, when using grey hydrogen, the relative impact (increased H_2 consumption, stack replacements) is small since the hydrogen production method itself is the main contributor to total GWP.

To confirm the relative importance of the operational regime and H_2 production method the share of specific phases (PEMFC manufacturing phase, H_2 production, and maintenance – stack replacements) in total GWP is presented in Fig. 8 for green hydrogen production scenarios. In dynamic operational regime stack replacements (maintenance) represents the biggest share in the case of EU Hydro scenario where stack replacements (5 times) accounts for 45% of the total GWP, followed by 43% for hydrogen production and "only" 11% for the manufacturing of the new PEMFC system. Contrary to hydro scenario, the dynamic operation in PV hydrogen production scenario represents the biggest share in total GWP (89%), with stack replacement accounting for 9%, and the manufacturing 2%. This also shows that extending the lifetime (improving durability) of the stacks can additionally decrease the environmental impacts, especially in the dynamic operational regime, because less stack replacements would be needed.

Analysing just hydro scenario, that is the scenario with the lowest absolute values of environmental impact indicators, it is obvious that the operational regime has the biggest influence (Fig. 9) and evident increase of all environmental impact indicators. On average, all environmental impacts, caused by degradation in the operational phase, increase by 51% for the steady-state and 248.9% for the dynamic operational regime.

Sensitivity analysis of technological improvements in PEMFC systems

Since the LCI model was verified with a bit outdated technological inventory data the sensitivity analysis of two main technological parameters was done. The platinum content (Pt loading) and the durability performance targets were varied

Table 4 – Absolute values of the environmental impact indicators per 1 kW h using different H_2 production, existing and new LCI model.

	Natural gas steam reforming			EU 28 Hydro power			EU 28 Wind Power			SI PV power		
	M+O (NG-SR)	M+O Steady-state (NG-SR)	M+O Dynamic (NG-SR)	M+O (Hydro)	M+O Steady-state (Hydro)	M+O Dynamic (Hydro)	M+O (Wind)	M+O Steady-state (Wind)	M+O Dynamic (Wind)	M+O (PV)	M+O Steady-state (PV)	M+O Dynamic (PV)
ADP elements [kg Sb eq.]	2.8E-03	4.6E-03	1.2E-02	5.1E-03	7.0E-03	1.4E-02	2.0E-02	2.2E-02	2.9E-02	1.3E-01	1.4E-01	1.4E-01
ADP fossil [MJ]	2.4E+05	2.5E+05	2.5E+05	2.3E+03	2.9E+03	4.8E+03	7.9E+03	8.7E+03	1.1E+04	5.0E+04	5.3E+04	5.5E+04
AP [kg SO_2 eq.]	7.5E+00	9.8E+00	1.8E+01	2.6E+00	4.6E+00	1.3E+01	3.8E+00	5.9E+00	1.4E+01	2.0E+01	2.3E+01	3.1E+01
EP [kg PO_4 eq.]	9.8E-01	1.2E+00	2.1E+00	2.9E-01	5.1E-01	1.4E+00	4.2E-01	6.4E-01	1.5E+00	1.6E+00	1.9E+00	2.7E+00
FAETP [kg DCB eq.]	9.0E+01	1.6E+02	4.4E+02	8.6E+01	1.6E+02	4.3E+02	9.0E+01	1.6E+02	4.4E+02	1.3E+02	2.0E+02	4.7E+02
GWP [kg CO_2 eq.]	1.4E+04	1.4E+04	1.5E+04	5.2E+02	6.2E+02	9.8E+02	7.4E+02	8.6E+02	1.2E+03	4.5E+03	4.7E+03	5.1E+03
GWP ex. Bio [kg CO_2 eq.]	1.4E+04	1.4E+04	1.5E+04	5.2E+02	6.2E+02	9.8E+02	7.3E+02	8.5E+02	1.2E+03	4.5E+03	4.8E+03	5.1E+03
HTP [kg DCB eq.]	1.2E+03	1.3E+03	1.8E+03	7.7E+02	9.1E+02	1.3E+03	1.3E+03	1.5E+03	1.9E+03	3.6E+03	3.8E+03	4.3E+03
MAETP [kg DCB eq.]	2.1E+05	3.8E+05	1.1E+06	2.2E+05	3.9E+05	1.1E+06	3.2E+05	4.9E+05	1.2E+06	1.8E+06	2.1E+06	2.7E+06
ODP [kg R11 eq.]	9.6E-07	1.7E-06	4.5E-06	9.6E-07	1.7E-06	4.5E-06	9.6E-07	1.7E-06	4.5E-06	9.8E-07	1.7E-06	4.5E-06
POCP, [kg Ethene eq.]	9.4E-01	1.1E+00	1.4E+00	1.2E-01	2.0E-01	5.3E-01	1.5E-01	2.4E-01	5.7E-01	1.7E+00	1.8E+00	2.2E+00
TETP [kg DCB eq.]	3.7E+00	4.1E+00	5.7E+00	3.8E+00	4.2E+00	5.8E+00	7.4E+00	8.0E+00	9.6E+00	2.8E+01	3.0E+01	3.1E+01

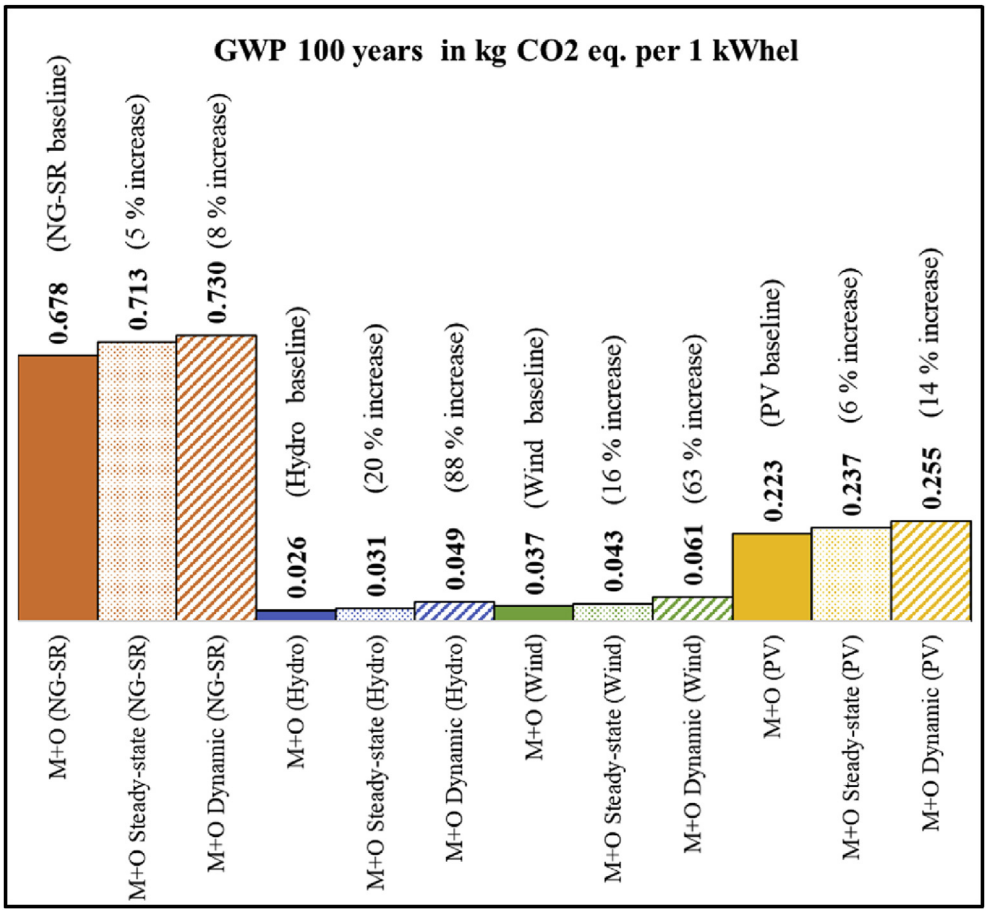


Fig. 7 – The influence of degradational effects and maintenance on environmental impacts of the 1 kW PEMFC system.

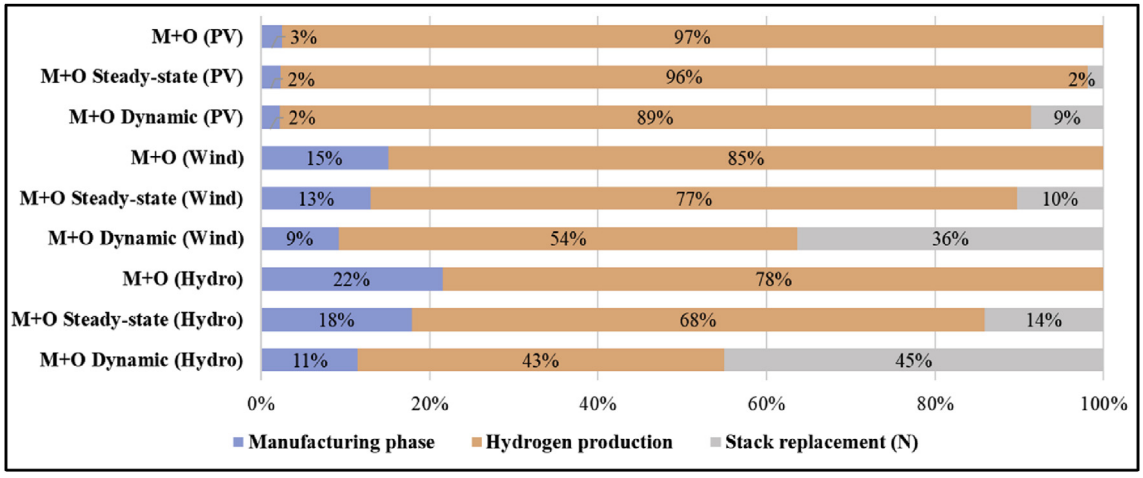


Fig. 8 – The share of the manufacturing phase, H_2 production method and maintenance (stack replacements) in total GWP for 1 kW h of PEMFC.

based on the future technological goals that are crucial in achieving greater commercialization, and consequently the competitiveness of hydrogen technologies in the energy sector [59]. The sensitivity analysis is performed for the considered scope of the LCA based on the green hydrogen production (EU Hydro, SI PV, and EU Wind) scenarios. For Pt loading the baseline scenario used is 0.75 g Pt per 1 kWel power of PEMFC and additional scenarios are 0.4 g (2020 goal)

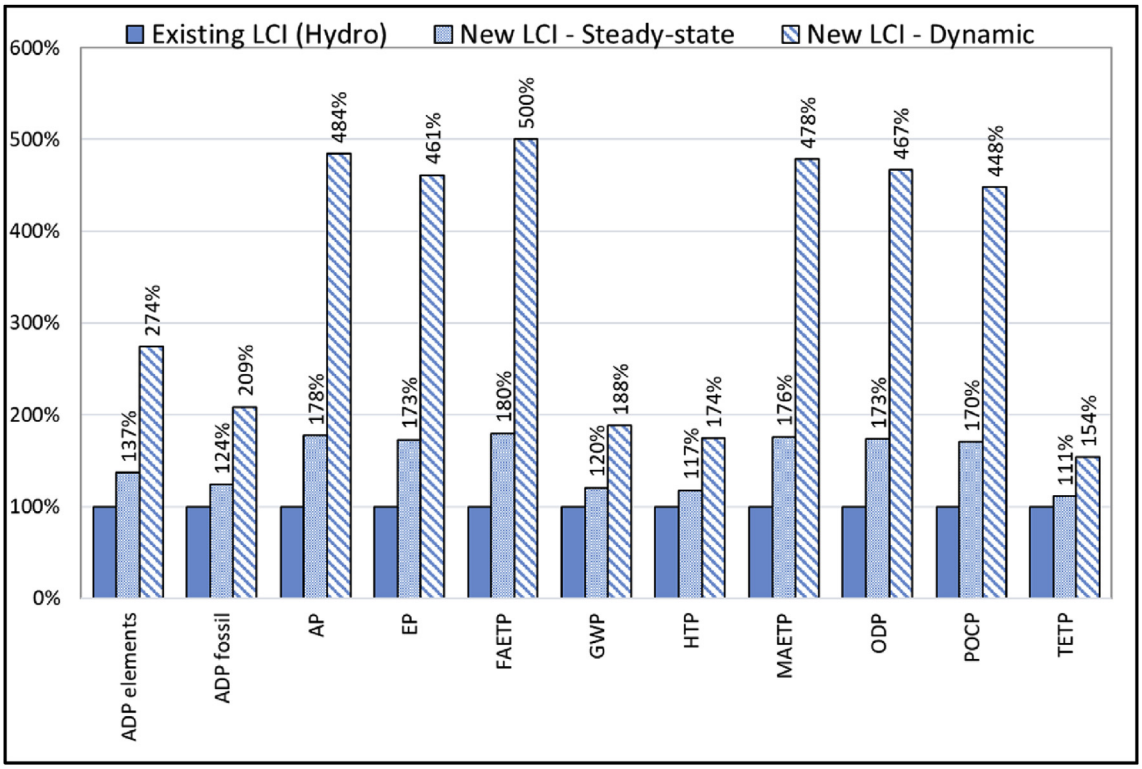


Fig. 9 – Relative increase of environmental impact indicators due to degradation processes in the case new LCI for a steady-state and dynamic operation for EU hydro scenario.

and 0.125 g (2030 goal) of Pt per 1 kWel power of PEMFC. For durability targets the reference scenario is $EoL_{dynamic-} = 3{,}800$ h (dynamic) and $EoL_{,steady\text{-}state} = 11{,}900$ h (steady state). The mid-term modelled scenario is $EoL_{dynamic-} = 5{,}000$ h and $EoL_{,steady\text{-}state} = 40{,}000$ h, and for long-term scenario $EoL_{dynamic} = 8{,}000$ h and $EoL_{,steady\text{-}state} = 80{,}000$ h.

The influence of lower Pt loading on environmental impacts

The Pt loading parameter is usually expressed as the amount of platinum per active surface area (mg/cm^2) of the MEA, but for the requirements of the LCA analysis the LCI data input was the total mass (anode + cathode side) of platinum per output power of the PEMFC. Platinum, which is used as a catalyst in MEA, belongs to the group of critical materials (CRM's2) [15], which have a high environmental impact at the phase of extraction of the raw materials, have a high price, and are geopolitically strategic raw materials. The goal of hydrogen technology manufacturers is to reduce the weight of platinum per kW output of PEMFC systems without compromising technological characteristics. Fig. 10 presents Pt loading impact on the GWP indicator for the dynamic operation (new LCI) of the 1 kW PEMFC system. For all Pt loadings (existing LCI and the new LCI) additional results are presented in the Supplementary material in S-Table 2 for existing LCI (without degradation impacts), in S-Table 3 for steady-state operation with new LCI, and in S-Table 4 for the dynamic operation with the use of new LCI model.

Platinum has the biggest environmental impacts in the manufacturing phase of the PEMFC system, even though only small masses are used. If we reduce the platinum content in the PEMFC stack, we can significantly reduce the environmental impacts of the PEMFC manufacturing phase and indirectly also the environmental impacts caused by replacements of degraded PEMFC stacks. The highest relative reduction of GWP, when the Pt loading is reduced from 0.75 g Pt per kW to 0.125 g Pt per kW, is 11% in EU hydro scenario, followed by 8.8% in EU Wind and 2.1% in SI PV scenario. The absolute value of GWP reduction in all scenarios is 107 kg CO_2 eq. for 1 kWe operating 20,000 h resulting in different relative reduction since each scenario has different absolute value of GWP for 20.000 kW h.

Analysing all environmental impact indicators (Fig. 11) for the EU hydro scenario we can see the average reduction in all environmental indicators by 72.6% for the 0.4 gPt/kW loading and 51.1% for 0.125 gPt/kW, in the case of dynamic operational regime. Platinum loading has a major impact in 7 out of 12 environmental indicators in CML2001. Based on these results, the development of new materials and/or the reduction of platinum content will play a very important role in reducing the overall environmental impact of hydrogen technologies. This will be even more visible in the case of green hydrogen production.

The influence of improved durability

In addition to Pt loading, different scenarios for PEMFC durability are analysed for manufacturing plus operational phase of the 1 kW PEMFC system. In the case of steady state, the operational lifetime of the PEMFC system is 11,900 h for the reference scenario (baseline case), 40,000 h for the mid-term scenario, and 80,000 h for the long-term scenario. In the case

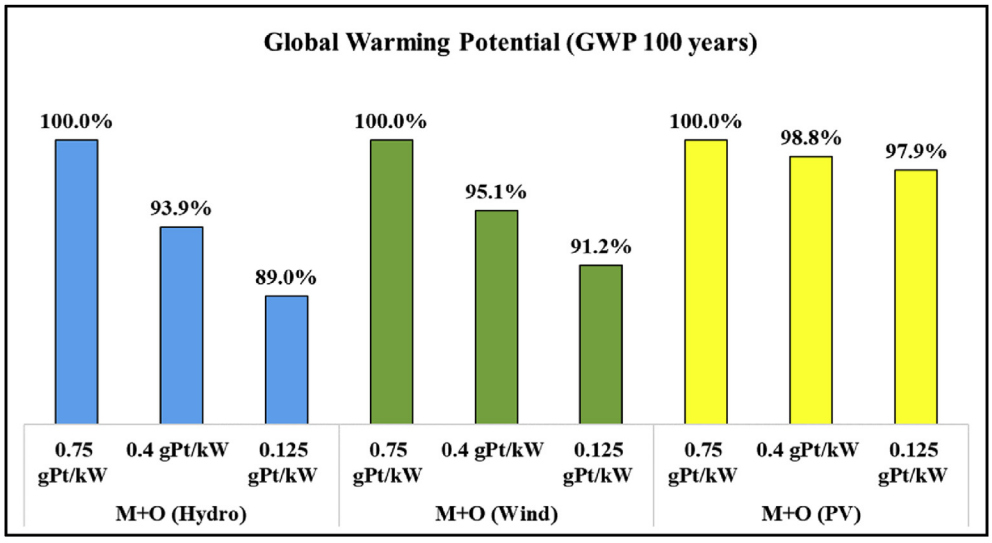


Fig. 10 – Relative reduction of GWP for the dynamic operational regime with reduced Pt loading in PEMFC stack for three scenarios.

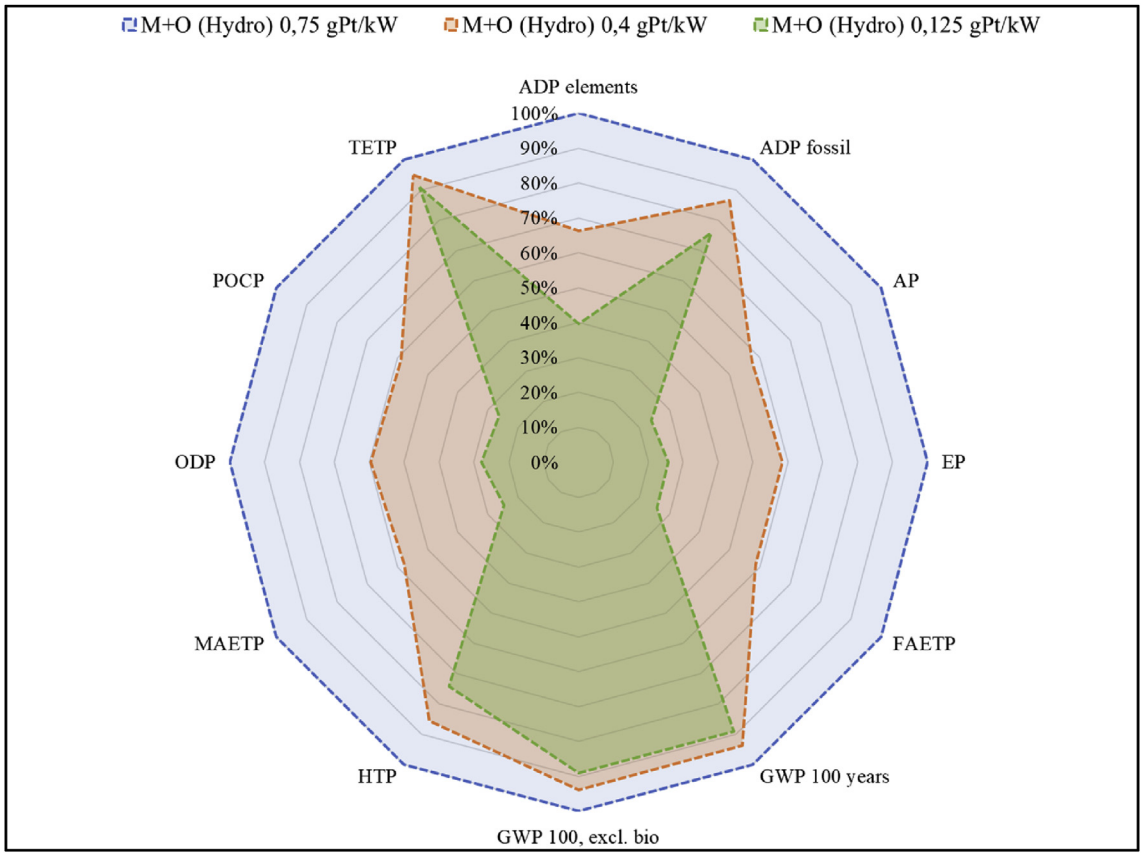


Fig. 11 – Relative comparison of environmental impact indicators for reduced Pt loadings in the case of dynamic PEMFC system operation and EU Hydro scenario.

of dynamic operation, the operational lifetime of the PEMFC system is 3,800 h for the reference scenario (baseline case), 5,000 h for the mid-term scenario, and 8,000 h for the long-term scenario. From the results of durability scenarios in Fig. 12 we can see that the GWP is greatly reduced with improved durability. Lower degradation in the operational lifetime of the PEMFC system (for dynamic and steady-state operational regime) increases the PEMFC efficiency and reduces the number of stack replacement needed. In the case of steady state operation and long-term durability, the GWP

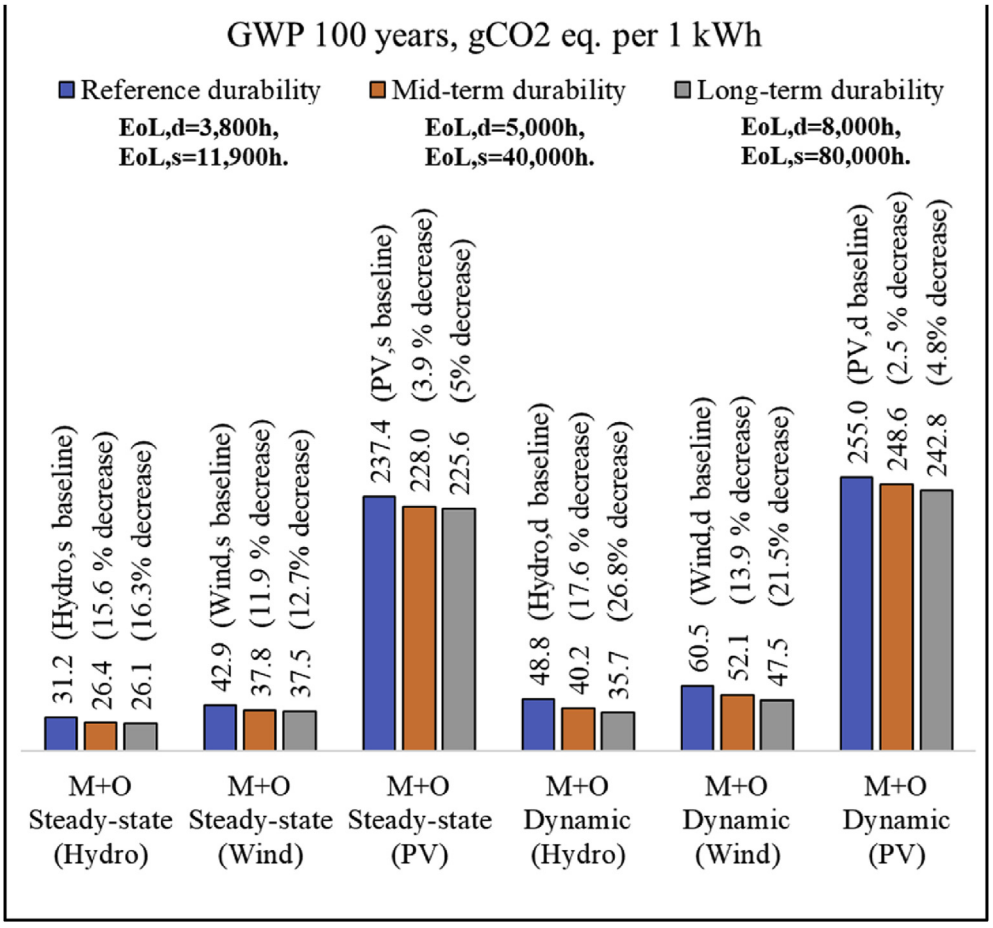


Fig. 12 – The influence of enhanced durability on GWP indicator for green hydrogen production scenarios. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

decreases for 11.8 $gCO_2eq.$ (5%) per generated 1 kWh in the case of PV hydrogen production scenario, 5.4 $gCO_2eq.$ (12.7%) per 1 kWh in the case of Wind hydrogen production scenario, and 5 $gCO_2eq.$ (16.3%) per 1 kWh in the case of Hydro hydrogen production scenario. According to durability targets for the steady state operation, the projected EoL operational time of the PEMFC stack is well beyond the lifetime used in the present study, which is 20,000 h. This means no stack replacements are needed which additionally decreases the environmental impacts. In the case of dynamic operational regime and improved durability the decrease in GWP emissions is even higher since the number of stack replacements is being significantly reduced (Fig. 12): from reference durability scenario $N_d = 5$, through mid-term durability scenario $N_d = 3$, and long-term durability scenario with $N_d = 2$. The highest relative GWP decrease is in the case of Hydro scenario 26.8%, followed by Wind with 21.5%, and PV scenario with 4.8%.

Conclusions

An innovative approach and development of a new life cycle inventory for the proton exchange membrane fuel cell is presented which includes the performance degradation for steady-state and dynamic operational regime. The semi-empirical model was developed using performance degradation data based on accelerated stress tests and integrated into LCI model. The efficiency drop and stack replacement are novel features in the new LCI model that directly or indirectly affect environmental impacts in operational and manufacturing phase. The LCI model was verified with a comparative 'cradle to the gate' life cycle assessment study of a 1 kWel PEMFC system operating for 20,000 h. Green hydrogen was used in the operational phase produced with hydro, wind, and photovoltaic electricity and benchmarked to natural gas reforming hydrogen. The influence of operational regime and degradational effects on environmental impacts was analysed using the new LCI model of the PEMFC system and compared to the environmental impacts of currently available LCI model. The highest relative increase in global warming potential was in the case of EU Hydro scenario, 88% for dynamic and 20% for steady state operational regime. This suggests that in the case of green hydrogen use the influence of operational regime is relatively more influential on total environmental impacts because every replacement of the PEMFC stack is added to environmental impacts due to additional manufacturing. This means when using green hydrogen in the future energy system based on renewable energy sources the

technological improvements (platinum loading, durability) in the manufacturing phase of the core components of PEMFC will become even more important. Sensitivity analysis of platinum loading showed that the reduction of 107 kg CO_2 eq. can be achieved in 20,000 h for 1 kWe PEMFC system operation with reducing the platinum loading from 0.75 g/kW to 0.125 g/kW. That represents 11% of GWP reduction in the case of green hydrogen scenario using hydroelectricity. If the durability of the PEMFC increases to 8,000 h for dynamic operational regime the GWP will decrease for 26.8%. For hydro, 21.5% for wind, and 4.8% for photovoltaic scenario.

The results of the new innovative LCI workflow presented in the paper provide an important contribution to the calculation and interpretation of the LCA of fuel cells and hydrogen technologies using PEMFC since degradation significantly affect the energy efficiency, leading to higher maintenance costs, stack replacements and higher hydrogen consumption in the operational phase. In addition, the newly developed LCI model can be easily updated with new data sets that correspond to technology improvements of PEMFC and integrated into LCA software using the LCI model presented in the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.ijhydene.2022.04.011.

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